Semester 3													
Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits	Examination Scheme					
							Theory			Practical		Total
							MS	IA	ES	LW	LE/Viva	
24INT151T	Civic and Social Service Internship	0	0	2	2	1	--	100	--	--	--	100
24MA201T	Mathematics - III	3	1	0	4	4	25	25	50	--	--	100
24CH202T	Introduction to Artificial Intelligence	3	0	0	3	3	25	25	50	--	--	100
24CH203T	Chemical Process Calculations	3	1	0	4	4	25	25	50	--	--	100
24CH204T	Mechanical Unit Operations	3	0	0	3	3	25	25	50	--	--	100
24CH204P	Mechanical Unit Operations Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH205T	Fluid Mechanics	3	0	0	3	3	25	25	50	--	--	100
24CH205P	Fluid Mechanics Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**15**	**2**	**6**	**23**	**20**						

Semester 4													
Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits	Examination Scheme					
							Theory			Practical		Total
							MS	IA	ES	LW	LE/Viva	
24INT251T	Industrial Orientation	0	0	0	0	0	--	--	--	--	--	Audit
OE-1	Open Elective 1 (Inter-School)	3	0	0	3	3	25	25	50	--	--	100
24CH206T	Chemical Engineering Industry 4.0	2	0	0	2	2	25	25	50	--	--	100
24CH206P	Chemical Engineering Industry 4.0 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH207T	Heat Transfer	3	0	0	3	3	25	25	50	--	--	100
24CH207P	Heat Transfer Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH208T	Chemical Process Technology	3	0	0	3	3	25	25	50	--	--	100
24CH208P	Chemical Process Technology Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH209T	Introduction to Numerical Methods for Chemical Engineers	3	1	0	4	4	25	25	50	--	--	100
24CH210T	Chemical Engineering Thermodynamics	3	0	0	3	3	25	25	50	--	--	100
24CH210P	Chemical Engineering Thermodynamics Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**17**	**1**	**8**	**26**	**22**						

24INT251T					Industrial Orientation					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	0	0	0	--	--	--	--	--	Audit (PP/NP)

COURSE OBJECTIVES

- To orient students to the working environment of a chemical process industry.
- To relate classroom learning to industrial practice through structured plant exposure.
- To familiarize students with plant safety culture, layout and utilities.
- To develop professional conduct and reporting skills.

COURSE CONTENT / ACTIVITIES

1. Pre-visit briefing on plant safety, personal protective equipment and code of conduct.
2. Structured industrial visit(s) / orientation programme at a chemical, petrochemical, energy or allied process plant.
3. Observation of plant layout, principal unit operations and unit processes, utilities and safety systems.
4. Interaction with practising engineers on process operation, control and maintenance practice.
5. Submission of an industrial orientation report and presentation before the departmental committee.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the organization and layout of a typical process plant.
CO2 - Understand the safety, health and environment practices followed in industry.
CO3 - Identify the unit operations and unit processes observed during the orientation.
CO4 - Relate the observed industrial practice to the concepts learnt in the classroom.
CO5 - Assess the role of a chemical engineer in plant operation and management.
CO6 - Communicate the observations effectively through a report and presentation.

TEXT / REFERENCE BOOKS

1. Pandit Deendayal Energy University. (2024). *Industrial orientation guidelines*. PDEU.
2. Austin, G. T. (1984). *Shreve's chemical process industries* (5th ed.). McGraw Hill.

EVALUATION

This is a mandatory zero-credit audit course. The award is PP (Passed) / NP (Not Passed) based on participation, the orientation report and the presentation.

24CH206T					Chemical Engineering Industry 4.0					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- To learn an overview of Industry 4.0.
- To acquire and fine-tune the skills and chemical engineering applications of data and machine learning.
- To gain an understanding of IoT and automation in chemical engineering.
- To understand recent developments in biomedical engineering.

Unit I: INTRODUCTION AND BASIC CONCEPTS OF INDUSTRY 4.0 **Hours: 6**

Introduction to Industry 4.0: definition, general framework, application areas; evolution of industrial revolutions; key features, need and benefits; Industry 4.0 core technologies - big data, advanced robotics, simulation, integration, Internet of Things (IoT), artificial intelligence (AI), cybersecurity, cloud computing, additive manufacturing and augmented reality; overview of machine learning for Industry 4.0; transformation of industrial processes through integration of modern technologies such as sensing and actuation, communication and computational processing.

Unit II: ROLE OF DATA AND MACHINE LEARNING FOR CHEMICAL ENGINEERS **Hours: 6**

Fundamentals of Python with emphasis on NumPy, Pandas, Matplotlib, OpenCV, TensorFlow and Keras; data handling; data visualization; logistic regression; fundamentals of artificial neural networks (ANN) and convolutional neural networks (CNN); use of data and machine learning in chemical and biochemical process industries; machine learning for chemical reactor design, for controlling chemical unit operations and for plant logistics.

Unit III: IoT, ROBOTICS AND AUTOMATION FOR CHEMICAL / BIOCHEMICAL ENGINEERS **Hours: 7**

Fundamentals of Arduino, ultrasonic sensors, temperature sensors and IR sensors and their applications in chemical and biochemical process industries; design of an IoT-enabled smart Air Quality Index (AQI) monitoring device; portable design of a weather monitoring unit deployable in a process plant; interfacing real-time data on the cloud for smart plant monitoring; IoT-enabled automated plant safety monitoring device; advanced process control in chemical industry; chemical plant simulator; development of mobile/web apps for smart plant monitoring using open-source resources.

Unit IV: BIOTECHNOLOGY AND BIOMEDICAL ENGINEERING **Hours: 7**

Targeted drug delivery; robotic surgery; 3D bioprinting; biomedical image analysis - MRI and ultrasound images; biosensors and their types; point-of-care testing devices; development of portable opto-electronic devices; role of self-propelled micro/nano motors in the healthcare sector; AI in healthcare with emphasis on POCT devices, patient health monitoring and telemedicine.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the fundamental concepts of Industry 4.0.
CO2 - Understand the basics of AI, augmented reality, IoT and cybersecurity.
CO3 - Understand the fundamentals of Python and useful libraries.
CO4 - Understand the use of data science and machine learning in chemical engineering.
CO5 - Understand the role of automation and IoT in chemical and biochemical engineering.
CO6 - Understand the concepts of advanced biomedical engineering.

TEXT / REFERENCE BOOKS

1. Raschka, S., & Mirjalili, V. (2019). *Python machine learning* (3rd ed.). Packt Publishing.
2. Goh, E. M. H. (2018). *Learn by examples: A quick guide to IoT with Arduino and data*. SV Book.
3. Shukla, A. N., & Quereshi, R. (2016). *Advanced biomedical engineering*. Discovery Publishing House.
4. Ustundag, A., & Cevikcan, E. (2018). *Industry 4.0: Managing the digital transformation*. Springer.

24CH206P					Chemical Engineering Industry 4.0 Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To learn practical aspects of Industry 4.0.
- To acquire an engineering grasp of Python coding.
- To gain an idea of open-source fluid flow simulations.
- To understand applications of machine learning for core engineers.

LIST OF EXPERIMENTS

1. Data analysis and data visualization using the Python programming language.
2. Implementation of a logistic regression problem in Python.
3. Implementation of an artificial neural network (ANN) using Python.
4. Development of a plant dashboard using Python.
5. Implementation of a convolutional neural network (CNN) for biomedical image analysis.
6. Solving fluid flow equations with Python programming.
7. Introduction to open-source CFD simulation software.
8. Introduction to ASPEN.
9. Arduino for portable spectrophotometric detection.
10. Arduino for water level monitoring using ultrasonic sound.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the concept of Industry 4.0 and its significance.
CO2 - Understand the resource requirements for the implementation of Industry 4.0.
CO3 - Learn the simulation packages for Industry 4.0.
CO4 - Explore the concept of fluid flow simulations.
CO5 - Inspect embedded platform applications for Industry 4.0.
CO6 - Synthesise the solution for a given Industry 4.0 related problem.

TEXT / REFERENCE BOOKS

1. Ustundag, A., & Cevikcan, E. (2018). *Industry 4.0: Managing the digital transformation*. Springer.
2. Kumar, K., Zindani, D., & Davim, J. P. (2019). *Digital manufacturing and assembly systems in Industry 4.0*. CRC Press/Taylor & Francis.
3. Patankar, S. V. (2017). *Numerical heat transfer and fluid flow*. CRC Press.
4. Raschka, S., & Mirjalili, V. (2019). *Python machine learning* (3rd ed.). Packt Publishing.
5. Goh, E. M. H. (2018). *Learn by examples: A quick guide to IoT with Arduino and data*. SV Book.

24CH207T					Heat Transfer					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the modes of heat transfer and their application in process industries.
- To understand heat balance equations in heat exchangers.
- To understand heat transfer with phase change.
- To understand heat transfer holistically, combining different modes of heat transfer.

Unit I: CONDUCTION BASED HEAT TRANSFER **Hours: 9**

Introduction and fundamentals of heat transfer; heat transfer rate, flux and resistances; thermal conductivity of materials; conduction through objects in series; insulation; unsteady state heat conduction; lumped heat capacity system; transient heat flow in a semi-infinite solid; thermal insulation; thermal safety considerations in high-conductivity materials.

Unit II: CONVECTION BASED HEAT TRANSFER **Hours: 12**

Film and overall heat transfer coefficients; solid-fluid heat transfer; natural and forced convection; laminar and turbulent flow heat transfer; nucleation and boiling; film-wise and drop-wise condensation; condenser design; fundamentals of pervaporation; calculation of the required cooling capacity for a given heat release rate; thermal safety considerations in convective heat removal processes.

Unit III: RADIATION BASED HEAT TRANSFER **Hours: 7**

Black and gray body radiation; Planck's law; Stefan-Boltzmann law; view factor; luminous and non-luminous gases; radiation hazards in industrial operations involving high-temperature equipment.

Unit IV: HEAT TRANSFER EQUIPMENT **Hours: 11**

Combined heat transfer; heat exchangers; coefficients for scale deposits; LMTD in heat exchangers; effectiveness; NTU method; finned tube heat exchangers; TEMA specifications-based analyses; thermal safety issues in heat exchanger and evaporator design.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the basic principles of heat transfer operations related to different modes of heat transfer.
CO2 - Analyse heat transfer operations involving combinations of heat transfer principles.
CO3 - Apply principles of heat transfer operations involving conduction, convection and radiation, including phase change.
CO4 - Analyse the problems related to heat transfer equipment.
CO5 - Understand industrial practices regarding heat transfer.
CO6 - Design heat transfer equipment based upon the conditions given in the problem statement.

TEXT / REFERENCE BOOKS

1. Coulson, J. M., & Richardson, J. F. (2000). *Chemical engineering* (Vols. 1-2, 6th ed.). Elsevier Press.
2. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill.
3. Holman, J. P., & Bhattacharyya, S. (2011). *Heat transfer* (10th ed.). McGraw Hill Education.
4. Dutta, B. K. (2014). *Heat transfer: Principles and applications*. PHI Learning.
5. Flynn, A. M., Akashige, T., & Theodore, L. (2019). *Kern's process heat transfer* (2nd ed.). Wiley-Scrivener.
6. Cengel, Y. A. (2004). *Heat transfer: A practical approach* (2nd ed.). McGraw-Hill.
7. Bergman, T. L., Lavine, A. S., Incropera, F. P., & DeWitt, D. P. (2011). *Incropera's principles of heat and mass transfer* (7th ed.). Wiley.

24CH207P					Heat Transfer Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To understand the modes of heat transfer (conduction, convection and radiation) and their application in process industries.
- To understand heat balance equations in heat exchangers.
- To understand heat transfer with phase change.
- To understand combined heat transfer involving all modes of heat transfer.

LIST OF EXPERIMENTS

1. Heat transfer in natural convection.
2. Heat transfer in forced convection.
3. Stefan-Boltzmann constant verification.
4. Parallel and counter flow heat exchangers.
5. Emissivity apparatus.
6. Thermal conductivity of insulating powder.
7. Thermal conductivity of a metal bar.
8. Heat transfer by vertical and horizontal heat exchangers in natural convection.
9. Plate type heat exchanger.
10. Finned tube heat exchanger.
11. Heat transfer in an agitated vessel.
12. Evaporator.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the basic principles of heat transfer operations related to different modes of heat transfer.
CO2 - Analyse heat transfer operations involving combinations of heat transfer principles.
CO3 - Apply principles of heat transfer operations involving conduction, convection and radiation, including phase change.
CO4 - Analyse the problems related to heat transfer equipment.
CO5 - Understand industrial practices regarding heat transfer.
CO6 - Design heat transfer equipment based upon the conditions given in the problem statement.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Heat transfer laboratory manual*. Pandit Deendayal Energy University.
2. Coulson, J. M., & Richardson, J. F. (2000). *Chemical engineering* (Vols. 1-2, 6th ed.). Elsevier Press.
3. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill.
4. Holman, J. P., & Bhattacharyya, S. (2011). *Heat transfer* (10th ed.). McGraw Hill Education.
5. Dutta, B. K. (2014). *Heat transfer: Principles and applications*. PHI Learning.
6. Cengel, Y. A. (2004). *Heat transfer: A practical approach* (2nd ed.). McGraw-Hill.

24CH208T					Chemical Process Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To provide a comprehensive overview of various organic and inorganic chemical manufacturing industries.
- To introduce the understanding of the processes, flow diagrams, process parameters and equipment utilized in chemical industries.
- To understand the properties, usage and application of intermediate chemicals and products.
- To bring an understanding of engineering problems in process manufacturing and to solve them by applying chemical engineering knowledge.

Unit I: INTRODUCTION AND RAW MATERIALS **Hours: 7**

History of organic and inorganic chemicals and their importance; structure of the chemical industry; raw materials and energy requirements (fossil fuels, oleochemicals and biomass, sulphuric acid, nitrogen, minerals); the role of sustainability; national and global trends in the chemical industry; raw materials and principles involved in producing olefins and aromatics; novel developments in olefin production technology; flash point, fire point and auto-ignition; effect of temperature and pressure on flash point.

Unit II: ORGANIC INDUSTRIES - 1 **Hours: 13**

Important derivatives from olefins and aromatics as building blocks: petrochemical derivatives of C1, C2, C3 and C4 olefin compounds and aromatics; safety aspects associated with specific industrial production (e.g. vinyl chloride, benzene derivatives).

Unit III: ORGANIC INDUSTRIES - 2 AND INORGANIC INDUSTRIES - 1 **Hours: 10**

Biotech industry (especially ethanol, biobutanol); biorefinery and processes for biomass conversion; production of useful chemicals from oleochemical feedstocks; biodiesel and biofuel production; glycerol valorisation; cellulosic derivatives using conventional and process intensification approaches. Inorganic chemicals: sulphuric acid, nitric acid, hydrochloric acid and phosphoric acid; safety aspects associated with exothermic or reactive chemical conversions in bio-refineries and acid production.

Unit IV: INORGANIC INDUSTRIES - 2 **Hours: 9**

Nitrogen industries: manufacture of ammonia, nitrogenous and mixed fertilizers; urea, ammonium sulphate, ammonium nitrate. Chlor-alkali industries: caustic soda and technological developments, chlorine, hydrochloric acid, soda ash and sodium bicarbonate; fire hazards and handling guidelines for chlorinated and nitrogen-based industrial products.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define, identify and select basic raw materials used for processes carried out in the chemical industry.
CO2 - Illustrate process flow diagrams / process block diagrams for the manufacture of various chemicals from the process description.
CO3 - Identify different unit operations and unit processes with their significance in the process.
CO4 - Analyse the technological options for problem solving in process industries.
CO5 - Recognize the importance of process economics in the industry.
CO6 - Discuss the impact of professional engineering solutions for societal benefits.

TEXT / REFERENCE BOOKS

1. Moulijn, J. A., Makkee, M., & van Diepen, A. E. (2013). *Chemical process technology* (2nd ed.). Wiley.
2. Mall, I. D. (2007). *Petrochemical process technology*. Macmillan India.
3. Gupta, O. P. (2020). *Elements of chemical process technology*. Khanna Publishing House.
4. Bhaskara Rao, B. K. (2007). *A textbook on petrochemicals*. Khanna Publishers.
5. Gopala Rao, M., & Sittig, M. (1997). *Dryden's outlines of chemical technology* (3rd ed.). East-West Press.
6. Austin, G. T. (1984). *Shreve's chemical process industries* (5th ed.). McGraw Hill.
7. Kirk, R. E., & Othmer, D. F. (1981). *Encyclopaedia of chemical technology* (3rd ed.). John Wiley.

24CH208P					Chemical Process Technology Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To find the practical relevance of chemical process technology theory.
- To illustrate the preparation methods of organic and inorganic compounds.
- To analyse the results of important compounds synthesised and purified.
- To explain the impact of hazardous chemicals on the environment and the safety required in chemical laboratories.

LIST OF EXPERIMENTS

1. Introduction to the laboratory, personal safety and material handling.
2. To prepare anthraquinone from anthracene.
3. Analysis of a Na2CO3 and NaHCO3 mixture.
4. To prepare soap in the laboratory and to carry out its cost analysis.
5. To evaluate the saponification value of a given oil sample.
6. To determine the acid value of given oil samples.
7. To prepare caustic soda by the chemical method.
8. Analysis of bleaching powder.
9. Determination of sodium carbonate in washing soda.
10. To prepare hydrated lime from given calcium carbonate powder.
11. Synthesis of biodiesel using edible/non-edible oil.
12. To estimate the fuel properties of prepared biodiesel.
13. To synthesise caustic soda by the electrochemical method.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Find the practical relevance of chemical process technology theory.
CO2 - Illustrate the preparation methods of organic and inorganic compounds involving reaction, separation and purification techniques.
CO3 - Characterize organic and inorganic compounds involving reaction, separation and purification techniques.
CO4 - Analyse the results related to important compounds synthesised and purified.
CO5 - Explain the impact of hazardous chemicals on the environment.
CO6 - Discuss the safety required for chemical laboratories.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Chemical process technology laboratory manual*. Pandit Deendayal Energy University.
2. Ray, S., & Das, G. (2020). *Process equipment and plant design: Principles and practices*. Elsevier Science.
3. Thakore, S. B., & Bhatt, B. I. (2018). *Introduction to process engineering and design* (2nd ed.). Tata McGraw Hill.
4. Sinnott, R. K. (2015). *Chemical engineering design: Principles, practice and economics of plant and process design* (2nd ed.). Butterworth Heinemann.
5. Austin, G. T. (1984). *Shreve's chemical process industries* (5th ed.). McGraw Hill.

24CH209T					Introduction to Numerical Methods for Chemical Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

Pre-requisite: Chemical Process Calculations; Mathematics - I and Mathematics - II

COURSE OBJECTIVES

- To introduce the numerical solutions of analytical problems.
- To understand the programmable methods to solve engineering problems.
- To implement the logical sequencing of numerical methods.
- To inculcate the numerical and analytical approach to solving problems.

Unit I: NUMERICAL SOLUTIONS TO ALGEBRAIC EQUATIONS **Hours: 13**

Mathematical modelling and chemical engineering problem solving; conservation laws; Taylor series and error propagation; bracketing methods - graphical methods, bisection method and false position method; open methods - fixed point iteration, Newton-Raphson, secant method; linear algebraic equations - Gauss elimination, LU decomposition, Gauss-Seidel.

Unit II: NUMERICAL APPROXIMATIONS AND DATA MANOEUVRING **Hours: 13**

Curve fitting - linear regression, multiple linear regression, polynomial regression, non-linear regression; interpolation; spline interpolation; Fourier series; numerical integration and differentiation - trapezoidal rule, Simpson's rules.

Unit III: NUMERICAL SOLUTIONS TO ORDINARY DIFFERENTIAL EQUATIONS **Hours: 13**

Ordinary differential equations - Euler method, Runge-Kutta 1, 2, 3 and 4 methods, adaptive RK methods, stiffness, boundary value and eigenvalue problems; chemical engineering problems: case studies.

Unit IV: NUMERICAL SOLUTIONS TO PARTIAL DIFFERENTIAL EQUATIONS **Hours: 13**

Partial differential equations - finite difference: elliptic equations, parabolic equations; finite element method - general approach, finite element in one dimension, two-dimensional problems; chemical engineering problems: case studies.

Max 52 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recognize the need for and efficacy parameters of numerical methods.
CO2 - Model chemical engineering problems as solvable by numerical methods.
CO3 - Implement numerical solutions in chemical engineering problem solving.
CO4 - Numerically solve algebraic equations and data manoeuvring problems.
CO5 - Solve ODE and PDE equations in chemical engineering through numerical methods.
CO6 - Design an independent solution strategy using various numerical methods.

TEXT / REFERENCE BOOKS

1. Chapra, S. C., & Canale, R. P. (2021). *Numerical methods for engineers* (8th ed.). McGraw-Hill Education.
2. Law, V. J. (2013). *Numerical methods for chemical engineers using Excel, VBA, and MATLAB*. CRC Press.
3. Kharab, A., & Guenther, R. B. (2023). *An introduction to numerical methods: A MATLAB approach* (5th ed.). Chapman and Hall/CRC.
4. Gilat, A. (2007). *Numerical methods for engineers and scientists: An introduction with applications using MATLAB*. John Wiley & Sons.
5. King, M. R., & Mody, N. A. (2010). *Numerical and statistical methods for bioengineering: Applications in MATLAB*. Cambridge University Press.

24CH210T					Chemical Engineering Thermodynamics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To provide a fundamental understanding of the various laws of thermodynamics.
- To develop an understanding of the PVT behaviour of a pure component.
- To derive various thermodynamic properties in terms of measurable variables.
- To provide a framework for the phase behaviour of pure components and mixtures.

Unit I: LAWS OF THERMODYNAMICS **Hours: 9**

Scope of chemical engineering thermodynamics; system and surroundings; quasi-static processes; diathermal and adiabatic boundaries; zeroth law and its implications; thermodynamic work; first law and its implications; heat; heat capacities; state of a pure component; closed thermodynamic system; statements of the second law; entropy of system and surroundings; spontaneity; first and second laws for open systems.

Unit II: VOLUMETRIC PROPERTIES OF PURE FLUIDS **Hours: 9**

General nature of PVT behaviour of pure substances; equations of state: ideal gas law; cubic equations of state - van der Waals, Redlich-Kwong, Soave-Redlich-Kwong and Peng-Robinson equations of state; generalized equations of state; the law of corresponding states; compressibility factor calculations using generalized correlation tables; virial equation of state; pressure–temperature relations and the associated safety aspects.

Unit III: THERMODYNAMIC PROPERTY RELATIONS AND INTRODUCTION TO VLE **Hours: 9**

Fundamental property relations; Maxwell's relations; enthalpy and entropy as functions of T and P; internal energy and entropy as functions of T and V; the Gibbs energy as a generating function; residual properties from various equations of state; generalized property correlations for gases; two-phase systems; Clapeyron equation; Antoine equation; Raoult's law; modified Raoult's law; bubble point and dew point calculations; vapour–liquid equilibrium; evaporation rate of a chemical after a leak using the Clausius–Clapeyron equation.

Unit IV: SOLUTION THERMODYNAMICS **Hours: 12**

Gibbs phase rule; chemical potential and the condition for phase equilibrium; partial properties; the ideal-gas-state mixture model; fugacity/activity and fugacity/activity coefficient calculations for pure species and mixtures; the ideal-solution model; excess properties; property changes of mixing; heat effects of mixing processes; chemical reaction equilibrium.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Outline the laws of thermodynamics.
CO2 - Understand the PVT behaviour of pure fluids.
CO3 - Apply different equations of state to find the molar volumes of gases, saturated vapours and liquids.
CO4 - Analyse the thermodynamic property relations in pure species and mixtures.
CO5 - Evaluate the residual properties from generalized correlations and equations of state.
CO6 - Solve vapour-liquid equilibrium problems such as bubble point, dew point and azeotropic concentration.

TEXT / REFERENCE BOOKS

1. Smith, J. M., Van Ness, H. C., & Abbott, M. M. (2018). *Introduction to chemical engineering thermodynamics* (8th ed.). McGraw-Hill.
2. Rao, Y. V. C. (1997). *Chemical engineering thermodynamics*. Universities Press.
3. Ott, J. B., & Boerio-Goates, J. (2000). *Chemical thermodynamics: Principles and applications*. Academic Press.
4. Prausnitz, J. M., Lichtenthaler, R. N., & de Azevedo, E. G. (1998). *Molecular thermodynamics of fluid-phase equilibria* (3rd ed.). Prentice Hall PTR.
5. Poling, B. E., Prausnitz, J. M., & O'Connell, J. P. (2000). *The properties of gases and liquids* (5th ed.). McGraw-Hill Professional.

24CH210P					Chemical Engineering Thermodynamics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To provide students with hands-on training on free and open-source tools related to the most common thermodynamic calculations in the chemical industries.
- To expose students to a numerical programming tool to solve complex thermodynamic calculations.
- To develop the ability to generate and interpret phase equilibrium data and diagrams.
- To apply iterative numerical methods to thermodynamic models.

LIST OF EXPERIMENTS

1. Understanding and creating cubic equations of state models in the Julia programming language.
2. Estimation of pure component enthalpy, entropy, residual enthalpy and residual entropy using cubic equations of state models.
3. Computation of Gibbs free energy, Helmholtz free energy, chemical potential and fugacity coefficients.
4. Computation of isothermal compressibility, isobaric expansivity, and isobaric and isochoric heat capacities.
5. Computation of mixing functions for a specified property.
6. Application of iterative methods to find solutions of cubic equations of state.
7. Dew point temperature and pressure estimation using the equation of state and activity coefficient models.
8. Bubble point temperature and pressure estimation using the equation of state and activity coefficient models.
9. Generating T-xy, P-xy and y-x diagrams for vapour-liquid equilibrium and prediction of the presence of azeotropes.
10. Generating and understanding the residue curve map for a multicomponent system.
11. Development of a flash drum for isothermal and adiabatic processes.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the cubic equation of state models.
CO2 - Understand the iterative methods to solve cubic equations of state.
CO3 - Determine the pure component properties and residual properties.
CO4 - Calculate the mixing functions to evaluate the mixture properties.
CO5 - Evaluate the bubble point and dew point properties using equation of state models.
CO6 - Develop the VLE data and various related diagrams.

TEXT / REFERENCE BOOKS

1. Prausnitz, J. M., Lichtenthaler, R. N., & de Azevedo, E. G. (1998). *Molecular thermodynamics of fluid-phase equilibria* (3rd ed.). Prentice Hall PTR.
2. Poling, B. E., Prausnitz, J. M., & O'Connell, J. P. (2000). *The properties of gases and liquids* (5th ed.). McGraw-Hill Professional.
3. Smith, J. M., Van Ness, H. C., & Abbott, M. M. (2018). *Introduction to chemical engineering thermodynamics* (8th ed.). McGraw-Hill.
4. Rao, Y. V. C. (1997). *Chemical engineering thermodynamics*. Universities Press.
5. Ott, J. B., & Boerio-Goates, J. (2000). *Chemical thermodynamics: Principles and applications*. Academic Press.